



FROM INTERNAL REFERENCE TO PERSISTENT CHANGE SIGNALS: HUMAN-IN-THE-LOOP 3D CHANGE DETECTION IN EXTENDED REALITY

Author

N.S. SURNAME

Student ID: 00000000

Supervised by

DR N.S. SURNAME

PROF. A. ANOTHER

A thesis submitted in fulfilment of the requirements for the degree of
Doctor of Philosophy

Department of Computer Science
University College London
2025

Doctoral Candidate Declaration

Replace this placeholder with the current UCL Doctoral Candidate Thesis Declaration Form before submission. UCL expects this declaration immediately after the title page, including declarations about your own work, contributions from others, use of AI, and inclusion of published works.

I confirm that the work presented in this thesis is my own unless otherwise stated. Contributions from collaborators, co-authors, supervisors, and others are declared in the relevant sections of this thesis and in the publications and author contributions statement.

Copyright Declaration

Replace this placeholder with the copyright and access statement required by UCL at submission. Confirm the wording against current UCL Doctoral School, Library Services, and UCL Discovery guidance before final deposit.

The copyright of this thesis rests with the author unless otherwise stated.

Where material has been reproduced from other sources, permission has been obtained where required and the source has been acknowledged.

Abstract

The abstract is a very brief summary of the dissertation's contents. It should be about half a page long.

- Audience: technical or academic peers.
- Content: concise summary of the problem, methodology, results, and conclusions.
- Language: uses discipline-specific terminology and assumes the reader has technical background knowledge.
- Purpose: to give experts a quick overview of the work so they can decide whether to read the full document.

Example of Abstract

We propose a novel convex optimization framework for parameter identification in non-linear dynamical systems using sparse measurements. The approach leverages structured regularization and is validated on a benchmark biochemical reaction model, showing improved accuracy and robustness compared to existing methods.

Plain Language Summary

Also this should be about half a page long.

- Audience: non-experts, such as policymakers, journalists, or the general public.
- Content: explains why the research matters, what was done, and what the implications and impact are, without technical jargon.
- Language: clear, simple, and accessible; often includes metaphors or examples to aid understanding. Do not use acronyms.
- Purpose: to communicate the significance and relevance of the work to a broad audience.

Example of Plain Language Summary (for the same project in the abstract!)

Can you see how different this is?)

Understanding how complex systems like biological reactions behave is important in medicine and engineering. We developed a new mathematical technique that can help uncover how these systems work, even if we only have limited data. This could help researchers make better predictions and design more effective treatments.

Impact Statement

This thesis investigates [briefly state the research area and problem]. Its potential impact lies in [describe the main academic, technical, societal, or practical beneficiaries].

Beyond the immediate research contributions, the work may support [describe future systems, methods, datasets, tools, standards, clinical/practical workflows, or research communities]. The thesis also identifies limitations and responsible-use considerations that should shape future deployment and evaluation.

Acknowledgments

It is usual to thank those individuals who have provided particularly useful assistance, technical or otherwise, during your project.

This is not needed, but common.

Publications and Author Contributions

This thesis incorporates material from the following publications, accepted manuscripts, or submitted manuscripts. Each relevant chapter also includes a publication note identifying the source material and summarising the author’s contribution.

Chapter	Publication	Status / venue	DOI / link
Chapter 2	Author list. “Paper 1 title.”	Published / accepted / submitted, venue, year.	DOI URL or
Chapter 3	Author list. “Paper 2 title.”	Published / accepted / submitted, venue, year.	DOI URL or
Chapter 4	Author list. “Paper 3 title.”	Published / accepted / submitted, venue, year.	DOI URL or
Chapter 5	Author list. “Paper 4 title.”	Published / accepted / submitted, venue, year.	DOI URL or

Author contributions

For each publication, state the contributions made by the doctoral candidate and by co-authors. This should include, where applicable, conceptualisation, study design, software implementation, data collection, experiment design, analysis, figure preparation, original drafting, and manuscript revision.

Paper 1. Replace this text with the specific author contribution statement for the publication used in Chapter 2.

Paper 2. Replace this text with the specific author contribution statement for the publication used in Chapter 3.

Paper 3. Replace this text with the specific author contribution statement for the publication used in Chapter 4.

Paper 4. Replace this text with the specific author contribution statement for the publication used in Chapter 5.

Use of published material

Where text, figures, tables, or data from the publications are reused, the relevant chapter cites the original source and indicates whether permission or an open licence permits reuse. If the copyright has been transferred to a publisher, replace or redact any material that cannot be included in the public deposit version of the thesis.

Contents

Doctoral Candidate Declaration	i
Copyright Declaration	ii
Abstract	iii
Plain Language Summary	iv
Impact Statement	v
Acknowledgments	vi
Publications and Author Contributions	vii
1 Introduction	1
1.1 Motivation	1
1.2 Research questions	1
1.3 Thesis contributions	1
1.4 Structure of the thesis	2
2 Paper-Based Chapter 1 Title	3
2.1 Chapter overview	3
2.2 Background and motivation	3
2.3 Method	4
2.4 Experiments and results	4
2.5 Discussion	4
2.6 Chapter summary and transition	4

3	Paper-Based Chapter 2 Title	5
3.1	Chapter overview	5
3.2	Background and motivation	6
3.3	Method	6
3.4	Experiments and results	6
3.5	Discussion	6
3.6	Chapter summary and transition	6
4	Paper-Based Chapter 3 Title	7
4.1	Chapter overview	7
4.2	Background and motivation	8
4.3	Method	8
4.4	Experiments and results	8
4.5	Discussion	8
4.6	Chapter summary and transition	8
5	Paper-Based Chapter 4 Title	9
5.1	Chapter overview	9
5.2	Background and motivation	10
5.3	Method	10
5.4	Experiments and results	10
5.5	Discussion	10
5.6	Chapter summary and transition	10
6	Synthesis and Discussion	11
6.1	Cross-chapter synthesis	11
6.2	Answering the research questions	11
6.3	Limitations	12
6.4	Future work	12
	Conclusions and future directions	13
A	Title of the Appendix	14

1

Introduction

Contents

1.1	Motivation	1
1.2	Research questions	1
1.3	Thesis contributions	1
1.4	Structure of the thesis	2

1.1 Motivation

Use this chapter to introduce the problem addressed by the thesis as a whole, rather than by any single paper. The opening should explain the practical or scientific setting, why the problem matters, and what gap remains despite existing work.

1.2 Research questions

State the thesis-level research questions here. For a thesis built from publications, these questions should make clear why the four paper chapters belong in one coherent doctoral project.

1.3 Thesis contributions

Summarise the main contributions of the thesis. This section should synthesise across the publications instead of repeating each paper abstract.

1.4 Structure of the thesis

Briefly explain how the thesis is organised. For example, Chapter 2 may introduce the first technical contribution, Chapter 3 may extend the method or evaluation, Chapter 4 may address a complementary challenge, and Chapter 5 may provide the final study or system contribution. Chapter 6 should then connect the findings across the four publications.

2

Paper-Based Chapter 1 Title

Contents

2.1 Chapter overview	3
2.2 Background and motivation	3
2.3 Method	4
2.4 Experiments and results	4
2.5 Discussion	4
2.6 Chapter summary and transition	4

Publication note. This chapter is based on: Author list, “Paper 1 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

2.1 Chapter overview

Do not repeat the paper abstract here. Instead, use this short overview to explain what problem this chapter addresses, how it advances the thesis-level research questions, and how it connects to the surrounding chapters.

2.2 Background and motivation

Summarise only the background needed for this chapter. Add thesis-specific context that may not have appeared in the original paper, especially where it helps the reader understand the wider doctoral project.

2.3 Method

Present the method, system, study design, or analytical approach. Rewrite the paper material into a consistent thesis voice and add details that were omitted from the publication because of page limits.

2.4 Experiments and results

Report the evaluation and findings. Reuse figures or tables only when copyright and permissions allow it, and cite the original publication wherever material is reused.

2.5 Discussion

Discuss what the results show, what they do not show, and how the chapter contributes to the thesis as a whole. This section can be broader than the paper discussion.

2.6 Chapter summary and transition

End with a short synthesis of the chapter's contribution and a transition to the next paper-based chapter.

3

Paper-Based Chapter 2 Title

Contents

3.1 Chapter overview	5
3.2 Background and motivation	6
3.3 Method	6
3.4 Experiments and results	6
3.5 Discussion	6
3.6 Chapter summary and transition	6

Publication note. This chapter is based on: Author list, “Paper 2 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

3.1 Chapter overview

Explain how this publication extends, contrasts with, or complements Chapter 2. The goal is to keep the thesis argument continuous rather than presenting four isolated papers.

3.2 Background and motivation

Add the context needed to make this chapter readable as part of the thesis.

3.3 Method

Present the chapter's technical, empirical, or design contribution.

3.4 Experiments and results

Report the evaluation and findings, adding any thesis-specific analysis that strengthens the argument.

3.5 Discussion

Explain the implications and limitations of the chapter.

3.6 Chapter summary and transition

Summarise the chapter and connect it to Chapter 4.

4

Paper-Based Chapter 3 Title

Contents

4.1	Chapter overview	7
4.2	Background and motivation	8
4.3	Method	8
4.4	Experiments and results	8
4.5	Discussion	8
4.6	Chapter summary and transition	8

Publication note. This chapter is based on: Author list, “Paper 3 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

4.1 Chapter overview

State the role of this paper within the thesis-level argument.

4.2 Background and motivation

Provide the specific context needed for this chapter.

4.3 Method

Describe the method, study, system, or analytical framework.

4.4 Experiments and results

Present the evidence and cite the original paper when reusing published material.

4.5 Discussion

Interpret the findings in relation to the whole thesis.

4.6 Chapter summary and transition

Summarise the contribution and connect it to Chapter 5.

5

Paper-Based Chapter 4 Title

Contents

5.1 Chapter overview	9
5.2 Background and motivation	10
5.3 Method	10
5.4 Experiments and results	10
5.5 Discussion	10
5.6 Chapter summary and transition	10

Publication note. This chapter is based on: Author list, “Paper 4 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

5.1 Chapter overview

Explain why this final paper-based chapter completes or extends the thesis argument.

5.2 Background and motivation

Add any context needed beyond the original publication.

5.3 Method

Describe the chapter's contribution.

5.4 Experiments and results

Present the evaluation and findings.

5.5 Discussion

Discuss the implications, limitations, and links to the earlier chapters.

5.6 Chapter summary and transition

Summarise the contribution and prepare the reader for the synthesis chapter.

6

Synthesis and Discussion

Contents

6.1	Cross-chapter synthesis	11
6.2	Answering the research questions	11
6.3	Limitations	12
6.4	Future work	12

6.1 Cross-chapter synthesis

Use this chapter to connect the four paper-based chapters into a single doctoral argument. The synthesis should make explicit what the combined body of work establishes that no single paper establishes alone.

6.2 Answering the research questions

Return to the research questions from Chapter 1 and answer them using evidence from Chapters 2–5.

6.3 Limitations

Discuss limitations across the whole thesis. This should include limitations inherited from individual papers and broader limitations that only become clear when the work is viewed as a combined programme of research.

6.4 Future work

Identify research directions that follow from the whole thesis, not only from the final paper.

Conclusions and future directions

All good projects conclude with an objective evaluation of the project's successes and failures and suggestions for future work which can take the project further. It is important to understand that there is no such thing as a perfect project.

Even the very best pieces of work have their limitations and you are expected to provide a proper critical appraisal of what you have done. Your assessors are bound to spot the limitations of your work and you are expected to be able to do the same.



Title of the Appendix

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.